both a sensor and a decision-maker, allowing swarms to perform complex coordinated missions such as surveillance, logistics, and combat operations with minimal oversight. This next-level swarm capability ensures adaptability even if communication links are disrupted.

- *Multi-domain operational capability*
  A defining feature of this generation is cross-domain adaptability, the ability to operate seamlessly in air, land, sea, and space environments. Through modular architecture and advanced propulsion systems, these drones can be deployed for aerial, maritime, or extraterrestrial exploration, expanding their strategic and scientific applications globally.
- *Advanced propulsion and energy systems*
  Fifth-generation UAVs utilize hybrid-electric, solar-assisted, and hydrogen-based propulsion systems, ensuring extended endurance and environmental sustainability. These systems enhance range, reduce acoustic signatures, and allow continuous operations for extended durations without refueling or maintenance interruptions.
- *Smart materials and self-healing structures*
  Built with next-generation composites and nanomaterials, these drones feature self-healing skins and adaptive morphing structures that can automatically repair minor damage and optimize aerodynamic profiles mid-flight. This innovation enhances survivability, reduces maintenance costs, and prolongs operational life cycles.
- *Digital twin integration and predictive maintenance*
  Each fifth-generation drone is paired with a digital twin, a virtual model that mirrors the drone’s condition, mission status, and system health in real time. This enables predictive diagnostics, performance